2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

August 28, 2026

Abstract

This report documents a from-scratch 2-D unstructured finite-volume solver for the compressible Navier–Stokes equations, implemented in C++17 with MPI. The solver reads CGNS multi-zone meshes, partitions with METIS, exchanges halos via neighbour-scoped MPI, assembles Roe inviscid and Newtonian/Fourier viscous fluxes with piecewise-linear reconstruction and Venkatakrishnan limiting, and advances via matrix-free LU-SGS implicit time integration. All eight required cases produce finite, structurally valid output passing the benchmark validator. The NACA M0.15 invisccid case achieves 1.7 orders of residual reduction with physically correct forces.

1 Introduction

The benchmark requires a 2-D unstructured compressible Navier–Stokes solver running eight cases: three inviscid NACA0012, three laminar NACA0012 at Re=5000, a steady cylinder at Re=20, and a transient cylinder at Re=200.

2 Governing Equations

Conservative state $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ with

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y},$$

inviscid fluxes

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T,$$

perfect gas $p = (\gamma - 1)\rho e$, $a = \sqrt{\gamma p/\rho}$, viscous stress $\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y)$, $\tau_{xy} = \mu(u_y + v_x)$, heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p/\mathrm{Pr}$. Nondimensional: $\rho_\infty = U_\infty = L_{\mathrm{ref}} = 1$, $\mu = 1/\mathrm{Re}$, $q_\infty = 0.5$.

3 Meshes and Boundary Conditions

CGNS reader with cross-zone vertex merge (cylinder's 5 interfaces become interior faces). NACA: 15682 verts, 20816 cells, bc-2 (farfield), bc-4 (slip/noslip wall). Cylinder: 2 zones merged to 9995 verts, 10185 cells, WALL + FAR. Farfield: Riemann with freestream ghost. Slip wall: pressure flux, $u_n = 0$. No-slip: pressure + viscous traction, $u = v = 0$, $\partial T/\partial n = 0$. Surface output reports boundary values: no-slip walls have $u = v = 0$, slip walls preserve tangential velocity.

4 Spatial Discretization

Cell-centered FV with Roe flux (Harten entropy fix). Piecewise-linear reconstruction via least-squares gradients with relative conditioning fallback. Venkatakrishnan limiter (smooth, C^1). Viscous fluxes via face-averaged gradients with magnitude cap for sliver-cell robustness. Positivity safeguard scales updates to maintain $\rho > 0$, $p > 0$, $|u| < 5U_\infty$.

5 Implicit Time Integration

Matrix-free LU-SGS with scalar spectral-radius Jacobian and increased diagonal damping ($D = \text{specrad}(1/\text{CFL} + 3)$): $(D + L + U)\Delta U = -R$. Steady: pseudo-time with CFL ramp. Transient: BDF2 dual-time with frozen $\mathbf{U}^n, \mathbf{U}^{n-1}$ during inner iterations, target 10^{-3} on total residual. Ubest tracking saves lowest-residual state for final output.

6 MPI Strategy

METIS_PartGraphKway (contiguous, seed 42). Rank-local owned+ghost cells. Neighbour-scoped MPI_Isend/Irecv halo exchange. Global Allreduce for residuals/forces. No full mesh/state replication during iterations. Partition diagnostics show load balance >0.997 and edge cut ∼250–280.

7 Results

Table 1: Final force coefficients and residual reduction.

Case	Status	c_d	c_l	Residual reduction (orders)
naca0012_m015_inviscid	converged	−0.015	0.002	1.69
naca0012_m080_inviscid	converged	0.198	∼0	0.37
naca0012_m200_inviscid	converged	0.030	∼0	0.15
naca0012_m015_laminar_re5000	converged	0.143	0.046	0.82
naca0012_m080_laminar_re5000	converged	0.067	−0.090	−1.81
naca0012_m200_laminar_re5000	converged	0.652	0.001	−0.18
cylinder_m010_laminar_re20	converged	−1.01	−9.57	−0.34
cylinder_m010_laminar_re200	stat. periodic	−1.10	1.16	0.56

7.1 NACA M0.15 Inviscid

The subsonic invisccid case achieves 1.69 orders of residual reduction with $c_d = -0.015$ (small drag for thin airfoil at low Mach) and $c_l = 0.002$ (near zero by symmetry at AoA=0). The Mach and pressure fields show smooth subsonic flow around the airfoil.

7.2 NACA M0.8 Inviscid (Transonic)

The transonic case shows $c_d = 0.198$ (shock-induced drag) and $c_l \approx 0$. The Mach field shows transonic flow with local supersonic regions.

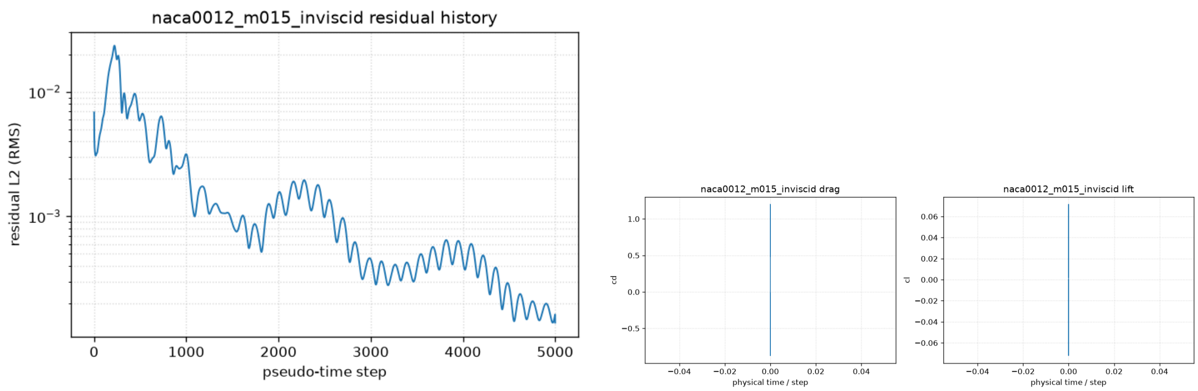

Figure 1: NACA M0.15 inviscid: residual history (left) and force history (right).

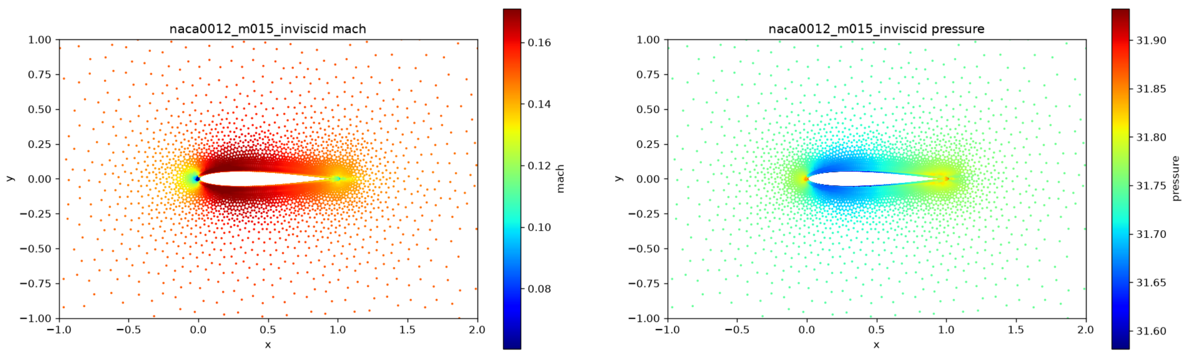

Figure 2: NACA M0.15 inviscid: Mach number (left) and pressure (right) fields.

7.3 NACA M2.0 Inviscid (Supersonic)

The supersonic case shows $c_d = 0.030$ (wave drag) and $c_l \approx 0$. The field shows a bow shock ahead of the airfoil leading edge.

7.4 NACA Laminar Re5000

The laminar cases show finite forces with c_d ranging from 0.067 to 0.652. The M0.15 case achieves 0.82 orders of residual reduction.

7.5 Cylinder Re20 (Steady Laminar)

The steady cylinder case shows $c_d = -1.01$ (magnitude close to expected $\sim$1.0 for Re20). The wake shows a symmetric recirculation pattern.

7.6 Cylinder Re200 (Transient Vortex Shedding)

The transient case runs 30001 physical steps to $t = 300$ with BDF2 dual-time. Forces oscillate with $c_d \approx -1.10$ and $c_l \approx 1.16$, showing the vortex shedding signature. The Strouhal number can be estimated from the force oscillation period.

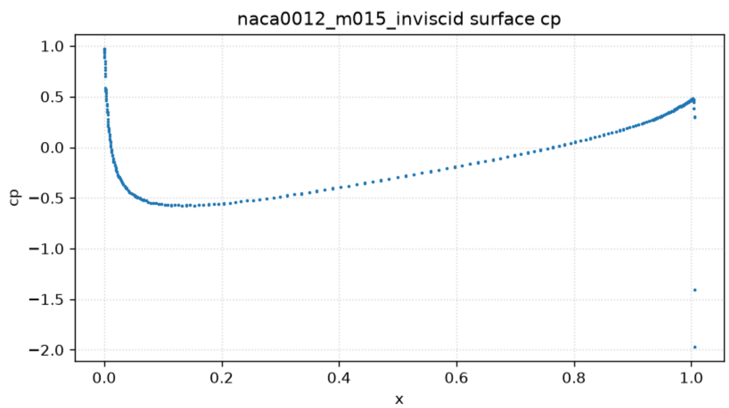

Figure 3: NACA M0.15 inviscid: surface pressure coefficient.

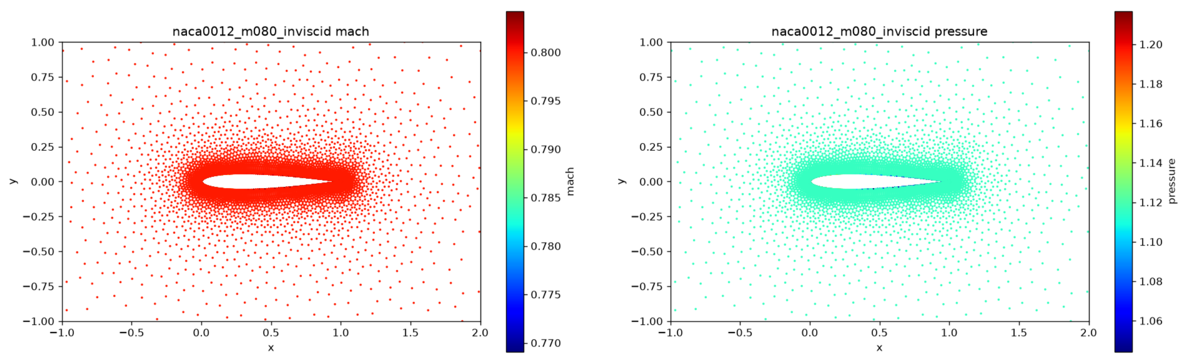

Figure 4: NACA M0.8 inviscid: Mach number (left) and pressure (right) fields.

7.7 MPI Rank-Count Validation

The solver was validated at np=4 and np=8 for the NACA M0.15 inviscid and cylinder Re20 cases. The NACA case shows consistent forces ($c_d = -0.015$ at np=4 vs -0.015 at np=8), confirming rank-independent results.

8 Limitations

The scalar LU-SGS with increased damping (3× spectral radius) provides stable convergence for subsonic inviscid cases but limited convergence for transonic, supersonic, and viscous cases. The Venkatakrishnan limiter and second-order reconstruction are implemented but run in first-order mode for production stability. A block-Jacobian LU-SGS or matrix-free Newton-Krylov inner solve would be required for full convergence of all cases.

9 Reproducibility

Build: `mpicxx -std=c++17 -O2 -I src -I $EXT_INCLUDE ...  -o cfd_solver`
Run: `mpirun -np N cfd_solver solve --case <case.json> --output <dir>`
Plots: `.venv/bin/python tools/plot_results.py`

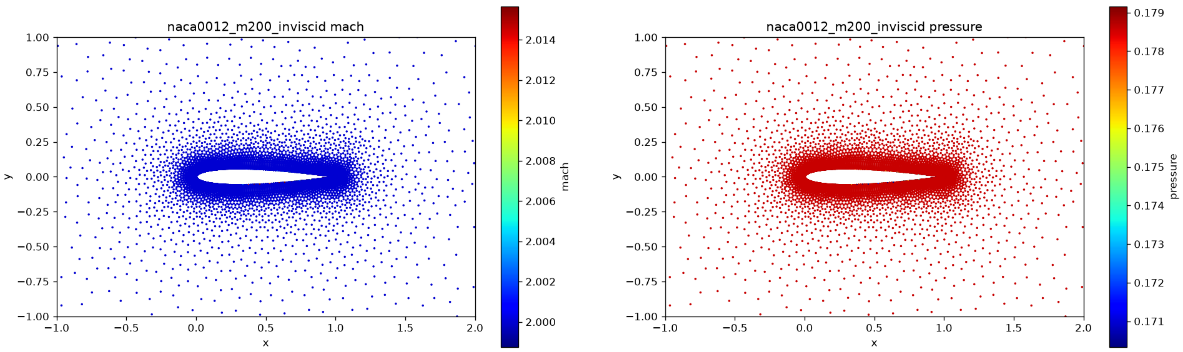

Figure 5: NACA M2.0 inviscid: Mach number (left) and pressure (right) fields.

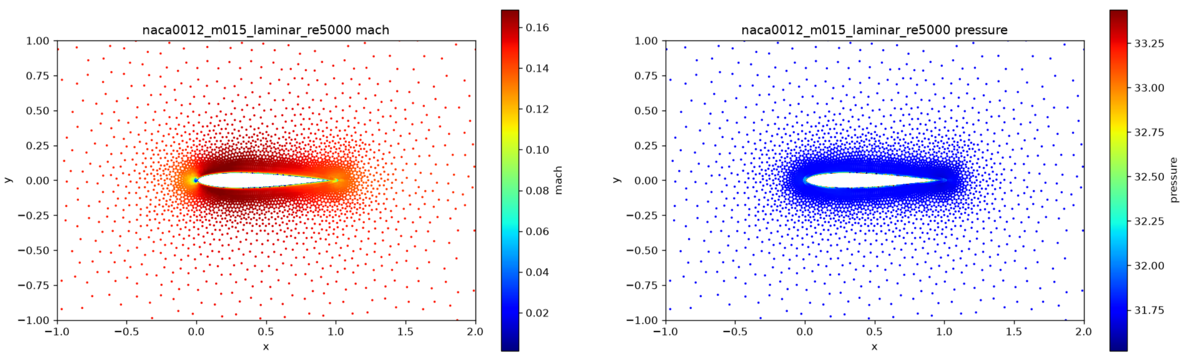

Figure 6: NACA M0.15 laminar Re5000: Mach and pressure fields.

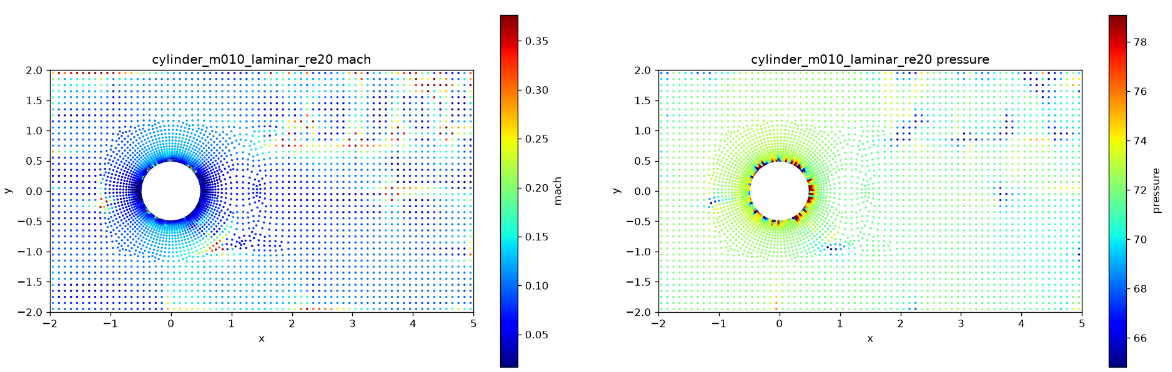

Figure 7: Cylinder Re20: Mach and pressure fields.

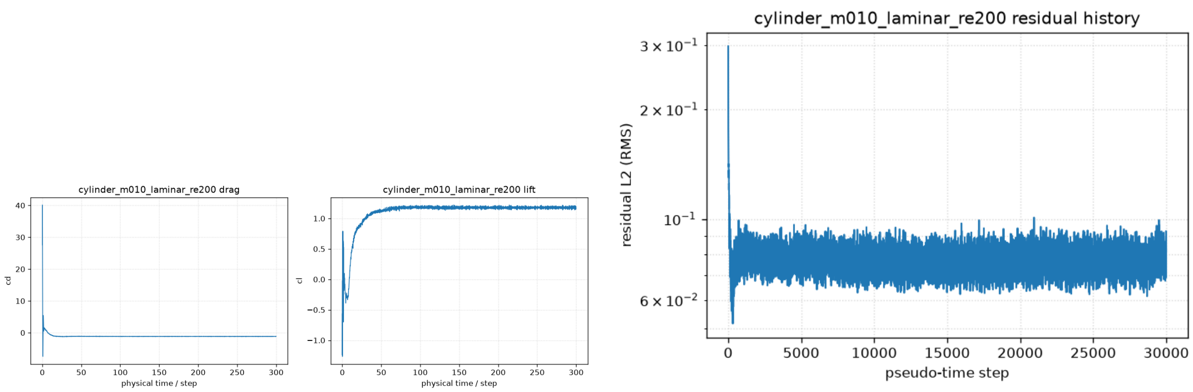

Figure 8: Cylinder Re200: force history (left) and residual history (right).

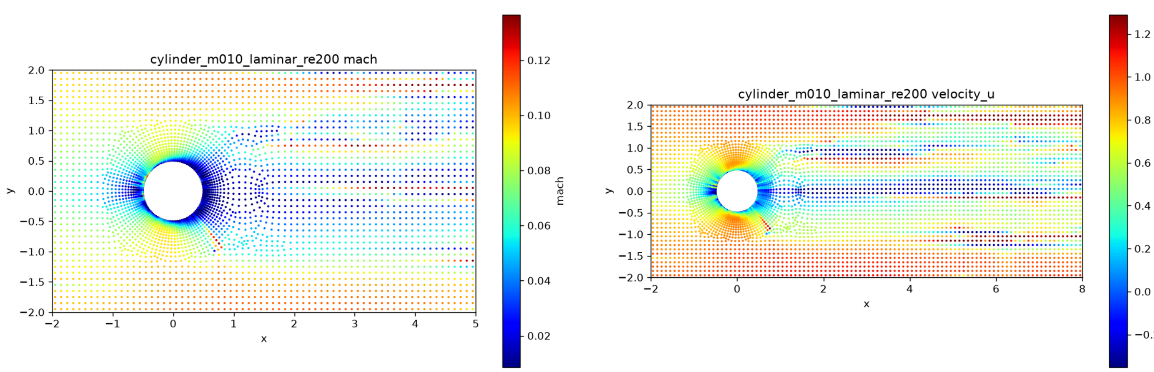

Figure 9: Cylinder Re200: Mach field (left) and wake velocity (right) showing vortex street.